

# Learner's Workbook

Worksheets and exercises for the 40-hour CDF intensive

SpaceCDF -- AI-Assisted Concurrent Design Facility

University of Ottawa - SEDTI

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**DRAFT**

## Learner's Workbook

### Worksheet 1.1: Introduction to Space Mission Design

|          |                                             |
|----------|---------------------------------------------|
| Name:    |                                             |
| Date:    |                                             |
| Team:    |                                             |
| Session: | 1.1 -- Introduction to Space Mission Design |

#### Part A: The 17 SE Processes -- SpaceCDF Mapping

For each NASA SE process (NPR 7123.1D), identify which SpaceCDF feature implements it. Write the tab name, button, or panel. If the process requires human judgment that the tool cannot automate, mark "Manual" in the last column.

Hint: Navigate through SpaceCDF's workflow steps (Need -> Concept -> Requirements -> Design) and explore each tab. Many processes map to more than one feature.

| #  | SE Process                          | SpaceCDF Feature(s) | Manual |
|----|-------------------------------------|---------------------|--------|
| 1  | Stakeholder Expectations Definition |                     |        |
| 2  | Technical Requirements Definition   |                     |        |
| 3  | Logical Decomposition               |                     |        |
| 4  | Design Solution Definition          |                     |        |
| 5  | Product Implementation              |                     |        |
| 6  | Product Integration                 |                     |        |
| 7  | Product Verification                |                     |        |
| 8  | Product Validation                  |                     |        |
| 9  | Product Transition                  |                     |        |
| 10 | Technical Planning                  |                     |        |
| 11 | Requirements Management             |                     |        |
| 12 | Interface Management                |                     |        |
| 13 | Technical Risk Management           |                     |        |
| 14 | Configuration Management            |                     |        |
| 15 | Technical Data Management           |                     |        |
| 16 | Technical Assessment                |                     |        |
| 17 | Decision Analysis                   |                     |        |

Discussion question: Which processes are well-supported by the tool? Which ones fundamentally require human judgment and cannot be fully automated? Write your answer below:

## Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase (Pre-A through F) it belongs to and which review gate marks the end of that phase:

| Activity                                                              | NASA  | Review Gate |
|-----------------------------------------------------------------------|-------|-------------|
| "We need to monitor sea ice thickness in the Canadian Arctic"         | _____ | _____       |
| Writing system-level "shall" requirements with verification methods   | _____ | _____       |
| Selecting a specific reaction wheel vendor after evaluating 3 options | _____ | _____       |
| Running thermal-vacuum qualification tests on the flight model        | _____ | _____       |
| Operating the satellite and producing daily data products             | _____ | _____       |
| Decommissioning the satellite and performing passivation              | _____ | _____       |
| Preliminary orbit trade study comparing SSO, ISS-orbit, and polar     | _____ | _____       |
| Detailed finite element analysis of the primary structure             | _____ | _____       |

Hint: Refer to the lifecycle phase table in Session 1.1, Section 4.3. Ask yourself: "What level of design detail does this activity require?"

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## Part C: CDF Position Interaction Analysis

For each design conflict below, identify which engineering positions must interact to resolve it. Then briefly describe how you would approach the resolution.

Conflict 1: The payload requires 50 W but the solar array can only provide 30 W.

Positions involved: \_\_\_\_\_

Resolution approach: \_\_\_\_\_

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Conflict 2: The reaction wheel vibration exceeds the payload's pointing stability requirement.

Positions involved: \_\_\_\_\_

Resolution approach: \_\_\_\_\_

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Conflict 3: The selected orbit (700 km) requires a propulsive deorbit for debris compliance, but there is no mass budget remaining for a propulsion system.

Positions involved: \_\_\_\_\_

Resolution approach: \_\_\_\_\_

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Conflict 4: The X-band downlink antenna pattern interferes with the payload's sensitive optical receiver.

Positions involved: \_\_\_\_\_

Resolution approach: \_\_\_\_\_

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## Part D: SpaceCDF Tool Exploration

Navigate through the SpaceCDF tool and answer each question:

1. How many KPI cards are shown on the Design Dashboard? List them by name:

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2. In the Gate Review tab, what are the MCR exit criteria? Which are auto-evaluated?

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3. What engineering positions are available when starting a collaborative session?

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4. What workflow steps does SpaceCDF guide you through (from left to right)?

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5. In the Positions tab, find your assigned position. What key questions does it list?

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## Part E: V-Model Sketch

In the space below, sketch the System-V model from memory. Label each level on both the left side (decomposition) and right side (integration). Draw the horizontal traceability arrows and label them.

Use this space for your sketch:

## Notes & Reflections

Record key points from the group discussion, questions you want to follow up on, and connections to your mission concept:

## Worksheet 1.2: Canadian Space Ecosystem & Regulatory Environment

|          |                                                          |
|----------|----------------------------------------------------------|
| Name:    |                                                          |
| Date:    |                                                          |
| Team:    |                                                          |
| Session: | 1.2 -- Canadian Space Ecosystem & Regulatory Environment |

## Part A: Canadian Space Industry Mapping

For your team's mission concept, identify potential Canadian industry partners at each tier. If no Canadian supplier exists for a capability, note what country you would source from and why.

| Tier                       | Capability Needed | Canadian Candidate(s) | International Alternative | Justification |
|----------------------------|-------------------|-----------------------|---------------------------|---------------|
| Tier 1 (Prime/Integration) |                   |                       |                           |               |
| Tier 2 (Subsystem)         |                   |                       |                           |               |
| Tier 2 (Subsystem)         |                   |                       |                           |               |
| Tier 3 (Component/Service) |                   |                       |                           |               |
| Tier 4 (R&D/Academic)      |                   |                       |                           |               |

Question: What CSA funding program(s) might support your mission? Why?

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## Part B: RSSSA Compliance Assessment

Answer the following questions for your team's mission concept:

- Does your mission involve any remote sensing of Earth (imaging, SAR, radiometry, etc.)?

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- If yes, will the satellite be operated from Canada?

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- What RSSSA conditions might apply? (Check all that apply)

| Condition                                      | Applies? | Notes |
|------------------------------------------------|----------|-------|
| Priority access for Government of Canada       |          |       |
| Shutter control provisions                     |          |       |
| Data disposition / handling plan               |          |       |
| System disposal plan                           |          |       |
| Raw data protection                            |          |       |
| Interoperability arrangements (foreign access) |          |       |

- If your mission does NOT involve remote sensing, what other regulatory requirements apply? (e.g., spectrum licensing, export control)

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## Part C: Spectrum Licensing Plan

Identify the frequency bands your mission will use and the licensing path for each:

| Link               | Band | Frequency (GHz) | Data Rate | Use (TT&C / Payload / ISL) | Licensing Path |
|--------------------|------|-----------------|-----------|----------------------------|----------------|
| Uplink             |      |                 |           |                            |                |
| Downlink (TT&C)    |      |                 |           |                            |                |
| Downlink (Payload) |      |                 |           |                            |                |
| Other              |      |                 |           |                            |                |

Hint: For a typical CubeSat, TT&C uses S-band (2.0--2.3 GHz) and payload data uses X-band (8.025--8.4 GHz). The licensing path for each is: ISED application -> ISED technical review -> ITU filing (API) -> ITU coordination -> ISED licence issuance.

Question: How early in your project should you begin the spectrum licensing process? Why?

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## Part D: Export Control Classification

List the 3 most critical components in your preliminary mission concept. For each, identify the likely export control classification:

| Component | Country of Origin | Likely Classification         | Licence Required? |
|-----------|-------------------|-------------------------------|-------------------|
|           |                   | ECCN: _____ / USML Cat: _____ | _____             |
|           |                   | ECCN: _____ / USML Cat: _____ | _____             |
|           |                   | ECCN: _____ / USML Cat: _____ | _____             |

Hint: Most commercial-off-the-shelf (COTS) space components from US manufacturers are classified under EAR (ECCN 9A515 for spacecraft parts). Military-grade or specifically designed items may fall under ITAR (USML Category XV). Canadian-origin components are subject to EIPA if they appear on the Export Control List.

Question: If your mission includes a US-origin star tracker and you plan to launch on an Indian PSLV, what export control issues arise?

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## Part E: Regulatory Timeline

Based on your mission concept, estimate when each regulatory activity should begin and how long it will take. Mark the activities on the timeline below:

| Regulatory Activity                 | Start Phase | Duration | Critical Path? |
|-------------------------------------|-------------|----------|----------------|
| RSSSA licence application           | _____       | _____    | _____          |
| ISED spectrum application           | _____       | _____    | _____          |
| ITU coordination                    | _____       | _____    | _____          |
| Export permit(s)                    | _____       | _____    | _____          |
| Launch provider interface agreement | _____       | _____    | _____          |

Question: Which regulatory activity is most likely to be on the critical path for your mission? Why?

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## Notes & Reflections

Record key points from the discussion, regulatory questions you need to research further, and implications for your mission design:

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## Worksheet 1.3: International Standards for Space

# Systems

|          |                                                  |
|----------|--------------------------------------------------|
| Name:    |                                                  |
| Date:    |                                                  |
| Team:    |                                                  |
| Session: | 1.3 -- International Standards for Space Systems |

## Part A: Standard Applicability Matrix

For your team's mission concept, determine which ECSS and NASA standards apply. For each standard, decide whether it is fully applicable, partially applicable (with tailoring), or not applicable. Justify your decision.

| Standard                                 | Branch | Applicability | Tailoring Notes / Justification |
|------------------------------------------|--------|---------------|---------------------------------|
| ECSS-M-ST-10C (Project Management)       | M      |               |                                 |
| ECSS-M-ST-40C (Configuration Management) | M      |               |                                 |
| ECSS-M-ST-80C (Risk Management)          | M      |               |                                 |
| ECSS-E-ST-10C (Systems Engineering)      | E      |               |                                 |
| ECSS-E-ST-20C (Electrical & Electronic)  | E      |               |                                 |
| ECSS-E-ST-32C (Structures)               | E      |               |                                 |
| ECSS-E-ST-40C (Software)                 | E      |               |                                 |
| ECSS-E-ST-50C (Communications)           | E      |               |                                 |
| ECSS-Q-ST-10C (Product Assurance)        | Q      |               |                                 |
| ECSS-U-AS-10C (Debris Mitigation)        | U      |               |                                 |
| ISO 24113 (Debris Mitigation)            | --     |               |                                 |
| NPR 7123.1D (SE Processes)               | --     |               |                                 |

Hint: A university CubeSat mission would typically invoke ECSS-E-ST-10C with heavy tailoring, ECSS-U-AS-10C fully, and might waive ECSS-Q-ST-10C (product assurance) due to cost constraints. A CSA-funded mission would typically require more complete ECSS compliance.

Question: What is the minimum set of standards your mission MUST comply with regardless of mission class?

## Part B: CDS Rev 14 Compliance Checklist

If your mission uses a CubeSat form factor, complete this compliance checklist. If not, skip to Part C.

| CDS Requirement                                    | Your Design Value | Compliant? | Notes |
|----------------------------------------------------|-------------------|------------|-------|
| Unit dimensions (100 x 100 x 113.5 mm per U)       | _____             | _____      | _____ |
| Mass per U ( $\leq 2.0$ kg)                        | _____             | _____      | _____ |
| Total mass (for your form factor)                  | _____             | _____      | _____ |
| Rail material (hard-anodised Al)                   | _____             | _____      | _____ |
| CG within 2 cm of geometric centre                 | _____             | _____      | _____ |
| No protrusions beyond envelope during launch       | _____             | _____      | _____ |
| RF silence for 30 min after deployment             | _____             | _____      | _____ |
| Deployment switches on all deployables             | _____             | _____      | _____ |
| Propulsion inhibits (3 independent, if applicable) | _____             | _____      | _____ |

Hint: Use SpaceCDF's Compliance panel to check CDS compliance. The tool will flag non-compliant parameters automatically based on your mass and dimensions entries.

## Part C: Debris Mitigation Compliance

Complete the following debris mitigation assessment for your mission orbit:

- Planned orbit altitude: \_\_\_\_\_ km
- Estimated natural orbital lifetime (without propulsion):

Use the rule of thumb:  $\tau_{\text{years}} \sim (h - 200)/(30) \times (m/A)/(50)$  where  $h$  is altitude (km),  $m$  is mass (kg),  $A$  is cross-section area ( $\text{m}^2$ )

$h =$  \_\_\_\_\_

$m/A =$  \_\_\_\_\_

$\tau =$  \_\_\_\_\_

- Compliance assessment:

| Requirement                       | Limit           | Your Mission | Compliant? |
|-----------------------------------|-----------------|--------------|------------|
| IADC post-mission lifetime        | $\leq 25$ years | _____        | _____      |
| FCC post-mission lifetime (2024+) | $\leq 5$ years  | _____        | _____      |
| ISO 24113 disposal reliability    | $\geq 90\%$     | _____        | _____      |

- If non-compliant with the 5-year FCC rule, what options exist?

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- Passivation plan (list all stored energy sources and how you will deplete them at end of life):

| Energy Source | Passivation Method |
|---------------|--------------------|
| Battery       | _____              |

|                           |  |
|---------------------------|--|
| Pressure vessels (if any) |  |
| Reaction wheels           |  |
| RF transmitter            |  |
| Other: _____              |  |

## Part D: Structural Design Margins

If your mission has preliminary structural design information, calculate the margin of safety:

Use the formula:  $MoS = (\sigma_{allowable}) / (FoS \times \sigma_{applied}) - 1$

Worked example: For Al 7075-T6 ( $\sigma_{yield} = 503$  MPa), applied stress = 200 MPa,  $FoS (yield) = 1.25$ :

$$MoS = (503) / (1.25 \times 200) - 1 = (503) / (250) - 1 = 1.012$$

This is a positive margin (compliant).

Your calculation:

Material: \_\_\_\_\_

$\sigma_{applied} =$  \_\_\_\_\_

$MoS =$  \_\_\_\_\_

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## Part E: SpaceCDF Compliance Panel

Navigate to the Compliance panel in SpaceCDF and answer:

1. What is the debris mitigation compliance status for your mission?

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2. What orbital altitude would be needed to comply with the FCC 5-year rule WITHOUT propulsion? (Use the tool's orbital lifetime calculator)

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3. What standards has the tool automatically suggested for your mission type?

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4. Are there any compliance flags (warnings or errors) on your current design?

## Notes & Reflections

Record key points about standards applicability, compliance challenges, and design implications for your mission:

## Worksheet 1.4: Mission Needs, Trade Studies & Concept of Operations

|          |                                              |
|----------|----------------------------------------------|
| Name:    |                                              |
| Date:    |                                              |
| Team:    |                                              |
| Session: | 1.4 -- Mission Needs, Trade Studies & ConOps |

### Part A: Stakeholder Analysis Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, priority,

and influence level.

| # | Stakeholder | Role | Key Needs | Constraints Imposed | Priority (P/S/T) | Influence (H/M/L) |
|---|-------------|------|-----------|---------------------|------------------|-------------------|
| 1 |             |      |           |                     |                  |                   |
| 2 |             |      |           |                     |                  |                   |
| 3 |             |      |           |                     |                  |                   |
| 4 |             |      |           |                     |                  |                   |
| 5 |             |      |           |                     |                  |                   |
| 6 |             |      |           |                     |                  |                   |

P = Primary (must satisfy), S = Secondary (should satisfy), T = Tertiary (nice to have)

H = High influence, M = Medium, L = Low

Question: Which stakeholder has the most influence over your mission? Do any stakeholders have conflicting needs? If so, how might you resolve the conflict?

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## Part B: Problem Statement

Write your team's mission problem statement. It must answer all four questions: What is the problem? Who is affected? What is the impact? What constraints exist?

Problem Statement:

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Peer Review Checklist (exchange with another team):

| Criterion                                        | Pass? | Feedback |
|--------------------------------------------------|-------|----------|
| Describes WHAT is needed, not HOW to achieve it? |       |          |
| Identifies the capability gap?                   |       |          |



|                                                                                 |       |       |
|---------------------------------------------------------------------------------|-------|-------|
| Names specific affected stakeholders?                                           | _____ | _____ |
| Quantifies the impact of not solving it?                                        | _____ | _____ |
| States constraints (budget, schedule, regulatory)?                              | _____ | _____ |
| Free of solution language? (No specific satellite type, orbit, component named) | _____ | _____ |

## Part C: Mission Objectives

Write 2--3 mission objectives. Score each against the quality checklist.

Objective 1: \_\_\_\_\_

| Quality Criterion                              | Yes/No | Notes |
|------------------------------------------------|--------|-------|
| States WHAT, not HOW?                          | _____  | _____ |
| Has measurable criterion with number + unit?   | _____  | _____ |
| Achievable with current technology (TRL >= 4)? | _____  | _____ |
| Relevant to the problem statement?             | _____  | _____ |
| Traceable to a specific stakeholder need?      | _____  | _____ |
| Classified as primary or secondary?            | _____  | _____ |

Objective 2: \_\_\_\_\_

| Quality Criterion                              | Yes/No | Notes |
|------------------------------------------------|--------|-------|
| States WHAT, not HOW?                          | _____  | _____ |
| Has measurable criterion with number + unit?   | _____  | _____ |
| Achievable with current technology (TRL >= 4)? | _____  | _____ |
| Relevant to the problem statement?             | _____  | _____ |
| Traceable to a specific stakeholder need?      | _____  | _____ |
| Classified as primary or secondary?            | _____  | _____ |

Objective 3 (optional): \_\_\_\_\_

## Part D: MoE / MoP / TPM Trace

For your primary objective, trace the measurement hierarchy:

Objective: \_\_\_\_\_

MoE (Measure of Effectiveness -- operational effectiveness from user perspective):

MoP (Measure of Performance -- technical performance of the system):

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TPM (Technical Performance Measure -- design parameter tracked during development):

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Hint: If your objective is "Provide 10 m GSD multispectral imagery...", then:

- MoE might be "% of target area with usable (<30% cloud) imagery within 5 days"
- MoP might be "GSD  $\leq$  10 m at nadir, SNR  $\geq$  100 in all bands"
- TPM might be "current estimate of aperture diameter (target: 15 cm), imager mass (target: 1.5 kg)"

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## Part E: Trade Study -- Space vs Non-Space

Complete a trade study evaluating whether a dedicated satellite is the right solution for your mission need.

Decision statement: \_\_\_\_\_

Alternatives:

| # | Alternative             | Description |
|---|-------------------------|-------------|
| 1 | _____                   | _____       |
| 2 | _____                   | _____       |
| 3 | _____                   | _____       |
| 4 | Do nothing / status quo | _____       |

Criteria and Weights (must sum to 1.0):

Use pairwise comparison or direct assignment. Show your weight derivation below.

| Criterion | Weight | Direction (higher/lower is better) |
|-----------|--------|------------------------------------|
| _____     | _____  | _____                              |
| _____     | _____  | _____                              |
| _____     | _____  | _____                              |
| _____     | _____  | _____                              |
| _____     | _____  | _____                              |
| Total     | 1.00   |                                    |

Weight derivation (pairwise comparison table or notes):

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Scoring Matrix (normalised 0--1):

| Alternative   | Criterion 1 | Criterion 2 | Criterion 3 | Criterion 4 | Criterion 5 | Weighted Total |
|---------------|-------------|-------------|-------------|-------------|-------------|----------------|
| 1: _____      | _____       | _____       | _____       | _____       | _____       | _____          |
| 2: _____      | _____       | _____       | _____       | _____       | _____       | _____          |
| 3: _____      | _____       | _____       | _____       | _____       | _____       | _____          |
| 4: Do nothing | _____       | _____       | _____       | _____       | _____       | _____          |

Recall: Weighted Total =  $\sum w_c \times s(a,c)$

Result: The highest-scoring alternative is: \_\_\_\_\_

Decision statement (1 paragraph -- include top 2 alternatives and why one was selected over the other):

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Sensitivity check: If you increase the weight of your second-most-important criterion by 20% (and re-normalise), does the ranking change?

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## Part F: Concept of Operations Outline

Document your preliminary ConOps following the NASA SEH Appendix S structure:

1. Mission Architecture:

Space segment: \_\_\_\_\_

Ground segment: \_\_\_\_\_

User segment: \_\_\_\_\_

2. Orbit-Average Power Budget:

Calculate using:  $P_{avg} = \sum P_{mode,i} \times DC_i$

| Mode | Power (W) | Duty Cycle (fraction of orbit) | P x DC (W) |
|------|-----------|--------------------------------|------------|
|------|-----------|--------------------------------|------------|

|                      |       |       |         |
|----------------------|-------|-------|---------|
| Idle / Housekeeping  | _____ | _____ | _____   |
| Science / Imaging    | _____ | _____ | _____   |
| Downlink             | _____ | _____ | _____   |
| Eclipse              | _____ | _____ | _____   |
| Orbit-average power: |       |       | _____ W |

Solar array requirement:  $P_{SA} =$  \_\_\_\_\_

Show your calculation:

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### 3. Data Budget:

Daily data generation: \_\_\_\_\_

Downlink capacity per pass: \_\_\_\_\_

Passes per day: \_\_\_\_\_

Daily downlink capacity: \_\_\_\_\_

Balance (downlink  $\geq$  generation)? \_\_\_\_\_

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## Part G: SpaceCDF Tool Entry

In SpaceCDF, enter your mission need (Step 1) and concept (Step 2). Then answer:

1. In Step 1, did the tool auto-detect your mission type from the objective text? What type did it suggest?

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2. In Step 2, run the mission trade analysis. Does the tool's recommendation match your manual trade study result from Part E?

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3. In the ConOps tab, build your architecture diagram. How many systems did you define?

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4. In the Trade Studies tab, run an orbit selection trade with at least 3 orbit options. What orbit did it recommend?

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## Notes & Reflections

Record key decisions made today, open questions, and what you want to investigate further:

## Worksheet 2.1: The System-V and Requirements Engineering

|                  |  |
|------------------|--|
| Name:            |  |
| Date:            |  |
| Team / Mission:  |  |
| Role / Position: |  |

## SpaceCDF Tab: Requirements

## Key Equations Reference

*Traceability Coverage:*

$$Coverage = \frac{N_{verified\ requirements}}{N_{total\ requirements}} \times 100\%$$

*Target: 100% -- every requirement needs a parent trace, a verification method, and a responsible subsystem.*

Requirement Quality (SMART):

| Specific -- Measurable -- Achievable -- Relevant -- Traceable

| Verification Methods (ATID):

| Analysis -- Test -- Inspection -- Demonstration

## Part A: System-V Mapping (10 min)

For your mission, identify one element at each level of the V-model:

| V-Level               | Left Branch (Decomposition) | Right Branch (Verification) |
|-----------------------|-----------------------------|-----------------------------|
| Stakeholder Need      |                             |                             |
| Mission Requirement   |                             |                             |
| System Requirement    |                             |                             |
| Subsystem Requirement |                             |                             |

## Part B: Requirement Quality Assessment (15 min)

Evaluate each requirement using the SMART checklist and NASA Appendix C criteria. Mark Pass (P) or Fail (F). Rewrite any failures.

| # | Requirement                                                   | S | M | A | R | T | WHAT or | Rewrite (if needed) |
|---|---------------------------------------------------------------|---|---|---|---|---|---------|---------------------|
| 1 | "The spacecraft shall operate at 500 km altitude"             |   |   |   |   |   |         |                     |
| 2 | "The system shall achieve GSD <= 10 m at nadir"               |   |   |   |   |   |         |                     |
| 3 | "The EPS shall use triple-junction GaAs cells"                |   |   |   |   |   |         |                     |
| 4 | "The system shall have a design lifetime of at least 3 years" |   |   |   |   |   |         |                     |
| 5 | "The structure shall be made of aluminium 7075"               |   |   |   |   |   |         |                     |

## Part C: Requirement Hierarchy and Traceability (15 min)

For your mission, write one requirement at each level. Draw the traceability chain.

Stakeholder Objective (traces FROM): \_\_\_\_\_

\_\_\_\_\_

\*\*Mission Requirement (MR- \_\_\_\_\_

\_\_\_\_\_

\*\*System Requirement (SR- \_\_\_\_\_

\_\_\_\_\_

\*\*Subsystem Requirement (SSR- \_\_\_\_\_

\_\_\_\_\_

Traceability diagram (draw arrows showing parent-child relationships):

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Part D: Splitting Compound Requirements (10 min)

Split this compound requirement into individually testable statements:

*"The system shall provide 10 m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."*

| # | Split Requirement (one concern each) | Type (F/P/I/C/E) | Verification (A/T/I/D) | Phase |
|---|--------------------------------------|------------------|------------------------|-------|
| 1 |                                      |                  |                        |       |
| 2 |                                      |                  |                        |       |
| 3 |                                      |                  |                        |       |
| 4 |                                      |                  |                        |       |
| 5 |                                      |                  |                        |       |

\_\_\_\_\_

\_\_\_\_\_



| Req ID           | Requirement Text (abbreviated) | Type (Functional / Performance / Interface / Constraint / Environmental) |                      |
|------------------|--------------------------------|--------------------------------------------------------------------------|----------------------|
| Req ID           | Action (Accept/Edit/Reject)    | Rationale                                                                |                      |
| Req ID           | Verification Method (A/T/I/D)  | Phase (B/C/D/E)                                                          | Rationale for Method |
| Name:            | _____                          |                                                                          |                      |
| Date:            | _____                          |                                                                          |                      |
| Team / Mission:  | _____                          |                                                                          |                      |
| Role / Position: | _____                          |                                                                          |                      |

## SpaceCDF Tab: Functions

## Key Equations Reference

*Function Types: Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support*

*Universal Functions (every spacecraft):*

- 1. Generate electrical power*
- 2. Maintain thermal environment*
- 3. Survive launch environment*
- 4. Communicate with ground (TTC)*
- 5. Dispose at end of life*

*Coverage Rule: Every leaf function must trace to at least one requirement. No gaps.*

*Allocation Rule: Multi-allocated functions create interfaces that must be explicitly managed.*

## Part A: Mission Function Tree (20 min)

Draw your mission's complete function tree. Use minimum 3 levels of decomposition. Include both mission-specific and universal functions.

Mission-specific functions:

F-001: \_\_\_\_\_

+ - - F-002: \_\_\_\_\_

```
+-- F-002a: _____  
+-- F-002b: _____  
+-- F-003: _____  
+-- F-003a: _____  
+-- F-004: _____  
+-- F-005: _____
```

Universal functions:

```
F-010: Generate electrical power  
+-- F-011: _____  
+-- F-012: _____  
F-020: Maintain thermal environment  
+-- F-021: _____  
F-030: Survive launch environment  
F-040: Communicate with ground (TTC)  
F-050: Dispose at end of life
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## Part B: Function-to-Requirement Traceability (15 min)

For each leaf function, write one derived requirement with a measurable threshold:

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## Part C: Multi-Allocation Analysis (10 min)

Identify at least one function that is allocated to more than one subsystem:

Function: \_\_\_\_\_

Subsystem A (primary): \_\_\_\_\_

Subsystem B (contributor): \_\_\_\_\_

Interface created between A and B: \_\_\_\_\_

Who "owns" this function? \_\_\_\_\_

What interface requirements are needed? \_\_\_\_\_

## Part D: Performance Criteria (10 min)

For 4 key functions, define quantitative performance criteria:

## Part E: SpaceCDF Coverage Check (15 min)

After entering functions in SpaceCDF:

1. How many total functions in your tree? \_\_\_\_\_
2. How many leaf functions? \_\_\_\_\_
3. How many leaf functions have NO linked requirements (amber badge)? \_\_\_\_\_
4. List the uncovered functions and derive requirements for them:
5. Are there any functions the tool generated that are NOT appropriate for your mission? List and explain why you removed them:

## Notes & Reflections

## Worksheet 2.3: Orbit Selection and Mission Architecture

| Function ID      | Function Name         | Allocated To (Subsystem) | Derived | Type   |
|------------------|-----------------------|--------------------------|---------|--------|
| Function         | Performance Criterion | Value                    | Unit    | Source |
| Function ID      | Function Name         | Derived Requirement      |         |        |
| Name:            |                       |                          |         |        |
| Date:            |                       |                          |         |        |
| Team / Mission:  |                       |                          |         |        |
| Role / Position: |                       |                          |         |        |

SpaceCDF Tabs: Mission Architecture, Orbit Trade Advisor

### Key Equations Reference

Orbital period:  $T = 2\pi \sqrt{a^3/u}$  where  $a = R_E + h$ ,  $u = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$ ,  $R_E = 6371 \text{ km}$

Orbital velocity:  $v = \sqrt{u/a}$

Eclipse fraction:  $f_{\text{ecl}} = (1/\pi) \arccos(\sqrt{a^2 - R_E^2}/a)$

Sun-synchronous inclination:  $\cos(i) = -\frac{\dot{\Omega} a^{7/2} R_E^2 J_2}{\sqrt{u}}$  where  $J_2 = 1.0826 \times 10^{-3}$

Hohmann  $\Delta V$ :  $\Delta V_1 = \sqrt{u/r_1}(\sqrt{2r_2/(r_1+r_2)} - 1)$

Swath width:  $W = 2h \tan(\theta)$

### Part A: Orbital Mechanics Calculations (20 min)

Your selected orbit: Altitude  $h =$  \_\_\_\_\_

Show all working for each calculation:

1. Semi-major axis:

$a = R_E + h = 6371 +$  \_\_\_\_\_

\_\_\_\_\_

2. Orbital period:

$T = 2\pi\sqrt{a^3/\mu} = 2\pi\sqrt{}$  \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

3. Orbital velocity:

$v = \sqrt{\mu/a} = \sqrt{3.986 \times 10^{14} /}$  \_\_\_\_\_

\_\_\_\_\_

4. Eclipse fraction (maximum):

$f = (1/\pi) \arccos(\sqrt{a^2 - R_E^2}/a) = (1/\pi) \arccos(\sqrt{1 -}) =$  \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

5. Sunlight and eclipse time per orbit:

$t_{\text{sun}} = T \times (1 - f) =$  \_\_\_\_\_

$t_{\text{ecl}} = T \times f =$  \_\_\_\_\_

6. Sun-synchronous inclination (if applicable):

$\cos(i) =$  \_\_\_\_\_

Does this match your selected inclination? Y / N

\_\_\_\_\_

\_\_\_\_\_

## Part B: Orbit Trade Matrix (10 min)

From SpaceCDF's Orbit Trade Advisor, record the top 3 candidates:

| Rank | Orbit Type | Alt (km) | Inc (deg) | GSD | Revisit (d) | Natural Lifetime (yr) | FCC 5-yr? | Score |
|------|------------|----------|-----------|-----|-------------|-----------------------|-----------|-------|
| 1    |            |          |           |     |             |                       |           |       |
| 2    |            |          |           |     |             |                       |           |       |
| 3    |            |          |           |     |             |                       |           |       |

Selected orbit: \_\_\_\_\_

## Part C: Debris Compliance (10 min)

For your selected orbit:

| Parameter                      | Value       |
|--------------------------------|-------------|
| Natural orbital lifetime       | _____ years |
| FCC 5-year compliant?          | Y / N       |
| IADC 25-year compliant?        | Y / N       |
| Propulsion needed for deorbit? | Y / N       |
| If yes, estimated deorbit D V  | _____ m/s   |
| Deorbit method (if needed)     | _____       |

Show D V calculation if propulsion needed:

## Part D: Downstream Cascade Analysis (10 min)

How does your orbit selection affect each subsystem? Fill in the impact:

| Subsystem     | Orbit Parameter That Matters          | Impact on Your Design               |
|---------------|---------------------------------------|-------------------------------------|
| Payload (GSD) | Altitude = _____ km                   | GSD = _____ m                       |
| EPS (eclipse) | Eclipse fraction = _____              | Eclipse duration = _____ min        |
| Comms (link)  | Slant range at 10 deg elev = _____ km | FSPL increase = _____ dB            |
| Thermal       | Eclipse/sunlight cycling              | Hot case / Cold case concern: _____ |
| AOCS          | Orbit rate = _____ deg/min            | Nadir tracking rate needed          |
| Propulsion    | Lifetime = _____ yr                   | Propulsion needed? Y / N            |
| Radiation     | TID estimate = _____ krad/yr          | Electronics class: _____            |

## Part E: Discussion (10 min)

What would happen to your mission if you moved the orbit 100 km higher?

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What would happen if you moved it 100 km lower?

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## Notes & Reflections

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## Worksheet 2.4: Mission Architecture -- Segments, Interfaces, and Budgets

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|                  |  |
|------------------|--|
| Name:            |  |
| Date:            |  |
| Team / Mission:  |  |
| Role / Position: |  |



## Part B: Power Budget by Mode (15 min)

| Subsystem      | Safe (W) | Idle (W) | Imaging (W) | Downlink (W) | Eclipse (W) |
|----------------|----------|----------|-------------|--------------|-------------|
| OBC            |          |          |             |              |             |
| AOCS           |          |          |             |              |             |
| Payload        |          |          |             |              |             |
| Comms (TX)     |          |          |             |              |             |
| Thermal        |          |          |             |              |             |
| Total          |          |          |             |              |             |
| Duty cycle (%) |          |          |             |              |             |

Orbit-average power calculation (show working):

$P_{avg} =$  \_\_\_\_\_

SA power required:

$P_{recharge} = \frac{P_{ecl} \times t_{ecl\_sun} \times \eta_{charge}}{(\dots \times \dots)} / (\dots \times 0.9) =$  \_\_\_\_\_

$P_{SA,EOL} = P_{peak} + P_{recharge} =$  \_\_\_\_\_

$P_{SA,BOL} = P_{SA,EOL} / (1 - 0.025)^n =$  \_\_\_\_\_

## Part C: Interface Identification (15 min)

List the 5 most critical interfaces in your design:

| # | Subsystem A | Subsystem B | Interface Types (M/E/T/D/I/R/O) | Key Concern |
|---|-------------|-------------|---------------------------------|-------------|
| 1 |             |             |                                 |             |
| 2 |             |             |                                 |             |
| 3 |             |             |                                 |             |
| 4 |             |             |                                 |             |
| 5 |             |             |                                 |             |

Write 2 formal interface requirements for your most critical pair:

Interface: \_\_\_\_\_

IR-001: \_\_\_\_\_

IR-002: \_\_\_\_\_

Verification method for IR-001: \_\_\_\_\_

\_\_\_\_\_

## Part D: Conflict Resolution (10 min)

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict description: \_\_\_\_\_

\_\_\_\_\_

Severity: Critical / Major / Minor

Resolution options considered:

| # | Option | Pros | Cons |
|---|--------|------|------|
| 1 |        |      |      |
| 2 |        |      |      |
| 3 |        |      |      |

Selected resolution: \_\_\_\_\_

Rationale: \_\_\_\_\_

\_\_\_\_\_

## Part E: Budget Health Check (10 min)

From the SpaceCDF Dashboard, record all KPIs:

| Budget             | Value                          | Margin   | Status (G/A/R) |
|--------------------|--------------------------------|----------|----------------|
| Mass               | _____ kg wet vs _____ kg alloc | _____ %  |                |
| Power (worst mode) | _____ W vs _____ W SA          | _____ %  |                |
| Link               | _____ dB margin                | >= 3 dB? |                |
| Cost               | _____ MEUR vs _____ ceiling    | _____ %  |                |
| Pointing           | _____ deg vs _____ deg req     | _____ %  |                |

Which budget is tightest? \_\_\_\_\_

What single design change would improve it? \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Notes & Reflections

## Worksheet 3.1: Power System and Thermal Control Design

|                  |  |
|------------------|--|
| Name:            |  |
| Date:            |  |
| Team / Mission:  |  |
| Role / Position: |  |

SpaceCDF Tabs: Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

## Key Equations Reference

$$SA\ EOL\ power: P_{SA,EOL} = P_{peak} + \frac{P_{ecl} \times t_{ecl\_sun} \times \eta_{charge}}{t_{ecl\_sun}}$$

SA BOL power:  $P_{SA,BOL} = \frac{P_{SA,EOL}(1 - \delta)^n}{\eta_{SA}}$  (delta = 0.025/yr for GaAs)

SA area:  $A = \frac{P_{BO} \cdot L_{eta\_cell}}{S \cdot \cos\theta} \cdot f_{pack}$  (eta = 0.295, S = 1361 W/m<sup>2</sup>, f<sub>pack</sub> = 0.85)

Battery capacity:  $C = \frac{P_{ecl} \times t_{ecl}}{DOD \times \eta_{discharge}}$  (DOD = 0.30,  $\eta = 0.95$ )

Thermal equilibrium:  $T = (\frac{Q_{abs} + Q_{int}}{\epsilon \sigma A_{rad}})^{1/4}$  &nbsp;&nbsp;  ( $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$ )

ECSS thermal margins: Operating +/-5 degC, Acceptance +/-10 degC, Qualification +/-15 degC

## Part A: Solar Array Sizing (20 min)

Show all calculations step by step.

1. Peak sunlight power demand:  $P_{peak} =$  \_\_\_\_\_

2. Eclipse power demand:  $P_{ecl} =$  \_\_\_\_\_

3. Eclipse and sunlight times (from Worksheet 2.3):

$t_{ecl} =$  \_\_\_\_\_

4. Recharge power:

$P_{recharge} = \frac{P_{ecl} \times t_{ecl} \times \eta_{charge}}{(1 \times 10^3) \times 0.9} =$  \_\_\_\_\_

5. SA EOL required:

$P_{SA,EOL} = P_{peak} + P_{recharge} =$  \_\_\_\_\_

6. EOL degradation factor (mission lifetime = \_\_\_\_\_

$(1 - 0.025)^{1000} =$  \_\_\_\_\_

7. SA BOL required:

$P_{SA,BOL} = \frac{P_{SA,EOL}}{(1 - 0.025)^{1000}} =$  \_\_\_\_\_

8. SA area:

$A = \frac{P_{SA,BOL}}{0.295 \times 1361 \times \cos(45^\circ) \times 0.85} =$  \_\_\_\_\_

9. SA type needed (compare to CubeSat reference values):

- [ ] Body-mounted only (1U: ~2 W, 3U: ~7 W, 6U: ~12 W)
- [ ] Single deployable (1U: ~4 W, 3U: ~15 W, 6U: ~30 W)
- [ ] Dual deployable (3U: ~25 W, 6U: ~48 W)

10. SA mass:

$$m_{SA} = A \times \sigma = \underline{\hspace{10cm}}$$

(body-mounted:  $\sigma = 2.5 \text{ kg/m}^2$ ; deployable:  $\sigma = 1.5 \text{ kg/m}^2$ )

## Part B: Battery Sizing (10 min)

Show all calculations:

1. Eclipse energy:

$$E = P_{ecl} \times \frac{t_{ecl}}{60} = \underline{\hspace{10cm}}$$

2. Battery capacity required:

$$C = (E) / (DOD \times \eta) = (\hspace{1cm}) / (0.30 \times 0.95) = (\hspace{1cm}) / (0.285) = \underline{\hspace{10cm}}$$

3. With 20% margin:  $C_{spec} = \underline{\hspace{10cm}}$

4. Battery mass:

$$m_{bat} = \frac{C_{spec}}{E_{specific}} = (\hspace{1cm}) / (150) = \underline{\hspace{10cm}}$$

5. Cycle life check:

Number of eclipses in mission =  $\underline{\hspace{10cm}}$

At DOD = 30%, Li-ion provides ~10,000 cycles. Is this sufficient? Y / N

## Part C: Thermal Analysis (15 min)

1. Identify hot case and cold case for your orbit:

| Parameter | Hot Case | Cold Case |
|-----------|----------|-----------|
|-----------|----------|-----------|

|                      |           |           |
|----------------------|-----------|-----------|
| Solar illumination   |           |           |
| Operational mode     |           |           |
| Internal dissipation |           |           |
| Coating condition    | BOL / EOL | BOL / EOL |
| Key concern          |           |           |

2. From SpaceCDF Dashboard, record thermal predictions:

| Component        | Min Predicted (degC) | Max Predicted (degC) | Operating Min (degC) | Operating Max (degC) |
|------------------|----------------------|----------------------|----------------------|----------------------|
| Payload / sensor |                      |                      |                      |                      |
| Battery          |                      |                      |                      |                      |
| OBC              |                      |                      |                      |                      |
| Transponder      |                      |                      |                      |                      |

3. Margin check (need  $\geq 5$  degC):

4. If any margin fails, what thermal control action would you take?

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## Part D: SpaceCDF Verification (15 min)

Compare your hand calculations to SpaceCDF values:

| Component               | Hot margin = Max_op - Max_pred | Cold margin = Min_pred - Min_op | Pass?      |
|-------------------------|--------------------------------|---------------------------------|------------|
| Parameter               | Hand Calculation               | SpaceCDF Value                  | Difference |
| SA power required (BOL) | _____ W                        | _____ W                         | _____ W    |
| Battery capacity        | _____ Wh                       | _____ Wh                        | _____ Wh   |
| Orbit-average power     | _____ W                        | _____ W                         | _____ W    |
| Eclipse duration        | _____ min                      | _____ min                       | _____ min  |

If there are significant differences, explain why:

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## Notes & Reflections

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## Worksheet 3.2: Attitude and Orbit Control System (AOCS)

|                  |  |
|------------------|--|
| Name:            |  |
| Date:            |  |
| Team / Mission:  |  |
| Role / Position: |  |

SpaceCDF Tabs: Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

### Key Equations Reference

Gravity gradient torque:  $T_{gg} = (3u)/(2a^3)|I_z - I_x|\sin(2\theta)$

Aerodynamic torque:  $T_{aero} = (1)/(2)\rho v^2 C_D A * d_{cp-cm}$

SRP torque:  $T_{SRP} = (S)/(c) A_s (1+q) * d_{sp-cm}$

Magnetic torque:  $T_{mag} = M \times B$

Momentum storage:  $H = T_{dist} \times t_{desat}/2$

Pointing error (RSS):  $\theta_{total} = \sqrt{\sum \theta_i^2}$

MTQ torque:  $T_{MTQ} = m_{dipole} \times B \times \sin(\alpha)$

### Part A: AOCS Architecture Selection (10 min)

Pointing requirement from your mission: \_\_\_\_\_

AOCS architecture selected (tick one):

- ☐ Passive magnetic (> 5 deg)
- ☐ Magnetorquers only (2--5 deg)

- [ ] Reaction wheels + MTQ (0.1--2 deg)
- [ ] Fine pointing: RW + star tracker + MTQ (< 0.1 deg)
- [ ] Very fine: RW + ST + gyro + MTQ (< 0.01 deg)

Justification for selection:

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Estimated AOCS mass: \_\_\_\_\_

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## Part B: Disturbance Torque Calculations (20 min)

Calculate at least 2 disturbance torques for your spacecraft and orbit. Show all working.

Spacecraft properties:

| Parameter                                  | Value | Unit              |
|--------------------------------------------|-------|-------------------|
| Mass                                       |       | kg                |
| Form factor                                |       | U                 |
| I <sub>z</sub> (largest MOI)               |       | kg m <sup>2</sup> |
| I <sub>x</sub> (smallest MOI)              |       | kg m <sup>2</sup> |
| Cross-section area (largest face)          |       | m <sup>2</sup>    |
| d <sub>cp-cm</sub> (CP-CM offset estimate) |       | m                 |
| Orbit altitude                             |       | km                |

1. Gravity gradient torque:

$$T_{gg} = (3 \times 3.986 \times 10^{14}) / (2 \times (r)^3) \times |I_z - I_x| =$$

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$$T_{gg} =$$

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2. Aerodynamic torque (rho at \_\_\_\_\_)

$$T_{aero} = (1)/(2) \times$$

---



---

$$T_{aero} =$$

---

3. (Optional) Solar radiation pressure torque:

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$T_{SRP} =$  \_\_\_\_\_

4. (Optional) Residual magnetic dipole torque ( $M \sim$  \_\_\_\_\_)

$T_{mag} =$  \_\_\_\_\_

Summary:

| Source             | Torque (N m) |
|--------------------|--------------|
| Gravity gradient   |              |
| Aerodynamic        |              |
| SRP                |              |
| Magnetic dipole    |              |
| Total (RSS or sum) |              |

Which disturbance dominates? \_\_\_\_\_

## Part C: Reaction Wheel Sizing (10 min)

1. Torque requirement ( $\geq 2 \times$  total disturbance):

$T_{RW,req} = 2 \times$  \_\_\_\_\_

2. Momentum storage (desaturation interval = 1 orbit = \_\_\_\_\_)

$H = T_{dist} \times \sqrt{\frac{1}{\text{desat}^2}} =$  \_\_\_\_\_

3. Selected reaction wheel:

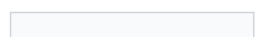
| Parameter | Required           | Selected Product |
|-----------|--------------------|------------------|
| Torque    | _____ mN m         | _____ mN m       |
| Momentum  | _____ mN m s       | _____ mN m s     |
| Mass      | --                 | _____ g          |
| Power     | --                 | _____ W          |
| Quantity  | 4 (3+1 redundancy) |                  |

## Part D: Pointing Error Budget (15 min)

Complete the RSS pointing error budget:

| Error Source    | Value (deg) | Value^2 |
|-----------------|-------------|---------|
| Sensor accuracy |             |         |
|                 |             |         |

Actuator resolution





|                     |            |  |
|---------------------|------------|--|
| Alignment knowledge |            |  |
| Thermal distortion  |            |  |
| Jitter (RW)         |            |  |
| Orbit knowledge     |            |  |
| Timing error        |            |  |
| Sum of squares      |            |  |
| RSS Total = sqrtsum | = ____ deg |  |

Requirement: \_\_\_\_\_

Margin: \_\_\_\_\_

Does the pointing budget close? Y / N

Which error source dominates? \_\_\_\_\_

What action would most improve the budget? \_\_\_\_\_

## Part E: SpaceCDF Verification (10 min)

Compare your hand calculations to SpaceCDF's Pointing Budget display:

| Parameter             | Hand Calculation | SpaceCDF | Match? |
|-----------------------|------------------|----------|--------|
| RSS pointing error    | ____ deg         | ____ deg |        |
| Dominant contributor  | _____            | _____    |        |
| Margin to requirement | ____ %           | ____ %   |        |

If there are differences, explain:

## Notes & Reflections

## Worksheet 3.3: Communications and Link Budget Design

|                  |  |
|------------------|--|
| Name:            |  |
| Date:            |  |
| Team / Mission:  |  |
| Role / Position: |  |

SpaceCDF Tabs: Link Budget, Spectrum Selector, Equipment Browser (Comms)

### Key Equations Reference

$$EIRP: EIRP = P_{TX} + G_{TX} - L_{TX} \text{ (dBW)}$$

$$FSPL: FSPL = 20 \log_{10}((4\pi d f)/(c)) \text{ (dB)}$$

$$G/T: G/T = G_{RX} - 10 \log_{10}(T_{sys}) \text{ (dB/K)}$$

$$C/N_0: C/N_0 = EIRP - FSPL - L_{losses} + G/T + 228.6 \text{ (dBHz)}$$

$$E_b/N_0: E_b/N_0 = C/N_0 - 10 \log_{10}(R_b) \text{ (dB)}$$

$$\text{Link margin: Margin} = E_b/N_{0,avail} - E_b/N_{0,req} - L_{impl} \geq 3 \text{ dB}$$

$$\text{Antenna gain: } G = \eta_a (\pi D / \lambda)^2$$

$$\text{dB conversion: } P_{dBW} = 10 \log_{10}(P_W); \text{ } 2 \text{ W} = +3.0 \text{ dBW}; 1 \text{ W} = 0 \text{ dBW}$$

### Part A: Frequency Band Selection (10 min)

Mission data rate requirement: \_\_\_\_\_

License type selected: Amateur / Experimental / Commercial

Band selected: \_\_\_\_\_

Rationale for band selection:

\_\_\_\_\_

\_\_\_\_\_

Licensing/filing required: \_\_\_\_\_

Estimated licensing cost: \_\_\_\_\_

\_\_\_\_\_

## Part B: Complete Link Budget (25 min)

Fill in ALL rows. Show computation for EIRP, FSPL, G/T, C/N<sub>0</sub>, and margin.

| Line | Parameter                                                 | Formula / Source                                        | Value  | Unit    |
|------|-----------------------------------------------------------|---------------------------------------------------------|--------|---------|
| 1    | TX Power                                                  | _____ W =                                               |        | dBW     |
| 2    | TX Antenna Gain                                           |                                                         |        | dBi     |
| 3    | TX Line Losses                                            |                                                         |        | dB      |
| 4    | EIRP = Line 1 + 2 + 3                                     |                                                         | _____  | dBW     |
| 5    | Free Space Path Loss                                      | $20\log_{10}(4\pi \times \text{ } \times \text{ } / c)$ |        | dB      |
| 6    | Atmospheric Loss                                          |                                                         |        | dB      |
| 7    | Pointing Loss                                             |                                                         |        | dB      |
| 8    | Polarisation Loss                                         |                                                         |        | dB      |
| 9    | RX Antenna Gain                                           |                                                         |        | dBi     |
| 10   | System Noise Temp                                         | _____ K =                                               |        | dBK     |
| 11   | G/T = Line 9 - 10                                         |                                                         | _____  | dB/K    |
| 12   | Boltzmann Constant                                        |                                                         | +228.6 | dBW/K/H |
| 13   | C/N <sub>0</sub> = 4 + 5 + 6 + 7 + 8 + 11 + 12            |                                                         | _____  | dBHz    |
| 14   | Data Rate                                                 | _____ bps = $10\log_{10}(\text{ } )$                    |        | dBbps   |
| 15   | E <sub>b</sub> /N <sub>0</sub> available = 13 - 14        |                                                         | _____  | dB      |
| 16   | E <sub>b</sub> /N <sub>0</sub> required (mod+code: _____) |                                                         |        | dB      |
| 17   | Implementation Loss                                       |                                                         |        | dB      |
| 18   | LINK MARGIN = 15 - 16 - 17                                |                                                         | _____  | dB      |

Show FSPL calculation:

$$\text{FSPL} = 20\log_{10}(4\pi) + 20\log_{10}(d) + 20\log_{10}(f) - 20\log_{10}(c)$$

$$= 21.98 + 20\log_{10}(\text{ } )$$

\_\_\_\_\_

\_\_\_\_\_

Does the link close ( $\geq 3$  dB)? Y / N

If margin is excessive, maximum achievable data rate at 3 dB margin:

$$R_{b,max} = 10^{(E_b/N_{0,avail} - E_b/N_{0,req} - L_{impl} - 3)/10} \times R_b = \underline{\hspace{10em}}$$


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## Part C: Data Budget (10 min)

Daily data generation:

$$V_{gen} = R_{payload} \times t_{imaging} \times N_{orbits} \times f_{compression}$$

$$= \underline{\hspace{10em}}$$

Daily downlink capacity:

$$V_{DL} = R_{DL} \times t_{contact} \times N_{passes} \times \eta_{protocol}$$

$$= \underline{\hspace{10em}}$$

Data budget closes? ( $V_{DL} \geq V_{gen}$ ): Y / N

If no, what is your mitigation plan?

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## Part D: SpaceCDF Comparison (10 min)

Open the Link Budget tab, enter your parameters, and compare:

| Parameter               | Hand Calculation | SpaceCDF | Difference |
|-------------------------|------------------|----------|------------|
| EIRP (dBW)              |                  |          |            |
| FSPL (dB)               |                  |          |            |
| C/N <sub>0</sub> (dBHz) |                  |          |            |
| Link Margin (dB)        |                  |          |            |

If there are differences, explain:

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## Part E: Modulation and Coding Selection (5 min)

Selected modulation: \_\_\_\_\_

Selected FEC code: \_\_\_\_\_

Required  $E_b/N_0$  at BER  $10^{-6}$ : \_\_\_\_\_

Spectral efficiency: \_\_\_\_\_

Rationale:

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## Notes & Reflections

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## Worksheet 3.4: Structure, Propulsion, and Equipment Selection

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|                  |  |
|------------------|--|
| Name:            |  |
| Date:            |  |
| Team / Mission:  |  |
| Role / Position: |  |



Axial launch load: \_\_\_\_\_

Load per rail (4 rails, equal sharing):

$$F_{\text{axial}} = \frac{m \times a_{\text{axial}} \times g_0}{4} = (\text{ } \times \text{ } \times 9.81) / (4) = \text{_____}$$

Stress in rail (cross-section  $8.5 \times 8.5 \text{ mm} = 7.225 \times 10^{-5} \text{ m}^2$ ):

$$\sigma = (F)/(A) = (\text{ })/(7.225 \times 10^{-5}) = \text{_____}$$

Margin of safety (yield, Al 7075-T6,  $\sigma_y = 503 \text{ MPa}$ ):

$$\text{MoS} = (503)/(\text{ } \times 1.25) - 1 = \text{_____}$$

Pass? Y / N

## Part C: Propulsion Decision (15 min)

Does your mission need propulsion?

| Question                                 | Answer      |
|------------------------------------------|-------------|
| Orbit altitude                           | _____ km    |
| Natural orbital lifetime                 | _____ years |
| FCC 5-year compliant without propulsion? | Y / N       |
| Orbit maintenance required?              | Y / N       |
| Constellation phasing required?          | Y / N       |
| Active deorbit required?                 | Y / N       |

Decision: &nbsp;   Propulsion / No propulsion

If propulsion is selected, complete this section. Show all working.

Total D V required:

| Manoeuvre                    | D V (m/s) |
|------------------------------|-----------|
| Orbit maintenance (_____ yr) |           |
| Deorbit                      |           |
| Collision avoidance margin   |           |
| Total                        |           |

Propulsion type selected: \_\_\_\_\_

Propellant mass (Tsiolkovsky):

$$m_{\text{prop}} = m_{\text{dry}} \times (e^{\Delta V / (I_{\text{sp}} \times g_0)} - 1)$$

= \_\_\_\_\_

System dry mass (thruster + tank + feed): \_\_\_\_\_

Total propulsion system mass: \_\_\_\_\_

Fraction of total spacecraft mass: \_\_\_\_\_

## Part D: Equipment Selection Log (25 min)

Using SpaceCDF's Equipment Browser, select components for each category:

| Category       | Component Selected | Manufacturer | Mass (kg) | Power | Cost (kEUR) | Qty | Total Mass |
|----------------|--------------------|--------------|-----------|-------|-------------|-----|------------|
| EPS Board      |                    |              |           |       |             | 1   |            |
| Battery        |                    |              |           |       |             | 1   |            |
| Solar Panels   |                    |              |           |       |             |     |            |
| OBC            |                    |              |           |       |             | 1   |            |
| Reaction Wheel |                    |              |           |       |             | x   |            |
| Magnetorquer   |                    |              |           |       |             | x   |            |
| Star Tracker   |                    |              |           |       |             |     |            |
| Sun Sensor     |                    |              |           |       |             | x   |            |
| Transponder    |                    |              |           |       |             | 1   |            |
| Antenna        |                    |              |           |       |             |     |            |
| Payload        |                    |              |           |       |             | 1   |            |
| Structure      |                    |              |           |       |             | 1   |            |
| Harness        |                    |              |           |       |             | 1   |            |
| Propulsion     |                    |              |           |       |             |     |            |
| TOTAL          |                    |              |           |       |             |     |            |

Parametric estimate (from Session 2.4): \_\_\_\_\_

Equipment-based total: \_\_\_\_\_

Difference: \_\_\_\_\_

Any incompatibilities detected? (RF band mismatch, voltage mismatch, interface conflict)

## Part E: Component Trade Study (15 min)

Subsystem traded: \_\_\_\_\_

| Criterion  | Weight | Option A: _____ | Score | Option B: _____ | Score | Option C: _____ | Score |
|------------|--------|-----------------|-------|-----------------|-------|-----------------|-------|
| Mass       | 0.20   |                 |       |                 |       |                 |       |
| Power      | 0.15   |                 |       |                 |       |                 |       |
| Cost       | 0.20   |                 |       |                 |       |                 |       |
| TRL        | 0.15   |                 |       |                 |       |                 |       |
| Heritage   | 0.15   |                 |       |                 |       |                 |       |
| Performanc | 0.15   |                 |       |                 |       |                 |       |
| Weighted   | 1.00   |                 |       |                 |       |                 |       |

Winner: \_\_\_\_\_

Rationale (beyond the numbers):

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## Part F: Final Budget Health Check (10 min)

After equipment selection, record final budget status:

| Budget                   | Value              | Margin   | Status    |
|--------------------------|--------------------|----------|-----------|
| Mass (wet vs allocation) | ____ / ____ kg     | ____ %   | G / A / R |
| Power (worst mode vs SA) | ____ / ____ W      | ____ %   | G / A / R |
| Link margin              | ____ dB            | >= 3 dB? | G / A / R |
| Cost vs ceiling          | ____ / ____ MEUR   | ____ %   | G / A / R |
| Pointing (RSS vs req)    | ____ / ____ deg    | ____ %   | G / A / R |
| Data (DL vs gen)         | ____ / ____ GB/day | ____ %   | G / A / R |

Any budget that does not close? \_\_\_\_\_

Proposed fix: \_\_\_\_\_

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## Notes & Reflections

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## Worksheet 4.1: Equipment Selection & Bill of Materials

Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

### Part A: Make/Buy/Reuse Decisions

For each subsystem, document the procurement decision and rationale:

| Subsystem                  | Decision (Buy/Reuse/Make) | Rationale | Lead Time |
|----------------------------|---------------------------|-----------|-----------|
| Structure                  |                           |           |           |
| EPS (solar array)          |                           |           |           |
| EPS (battery)              |                           |           |           |
| EPS (distribution)         |                           |           |           |
| OBC                        |                           |           |           |
| AOCS (sensors)             |                           |           |           |
| AOCS (actuators)           |                           |           |           |
| TTC (transceiver)          |                           |           |           |
| TTC (antenna)              |                           |           |           |
| Thermal                    |                           |           |           |
| Payload                    |                           |           |           |
| Propulsion (if applicable) |                           |           |           |

### Part B: Bill of Materials (BOM)

Complete the BOM from the SpaceCDF Equipment Browser selections:

### Part C: Budget Tracking

After all selections, record the live budget values from the SpaceCDF Dashboard:

| Item ID | Component | Manufacturer | Part Number | Qty | Mass (g) | Power (W) | Cost (kEUR) | TRL | ECCN |
|---------|-----------|--------------|-------------|-----|----------|-----------|-------------|-----|------|
| STR-001 |           |              |             |     |          |           |             |     |      |
| EPS-001 |           |              |             |     |          |           |             |     |      |
| EPS-002 |           |              |             |     |          |           |             |     |      |
| EPS-003 |           |              |             |     |          |           |             |     |      |

|          |            |                     |            |         |        |  |  |  |  |
|----------|------------|---------------------|------------|---------|--------|--|--|--|--|
| OBC-001  |            |                     |            |         |        |  |  |  |  |
| AOCS-00  |            |                     |            |         |        |  |  |  |  |
| AOCS-00  |            |                     |            |         |        |  |  |  |  |
| AOCS-00  |            |                     |            |         |        |  |  |  |  |
| TTC-001  |            |                     |            |         |        |  |  |  |  |
| TTC-002  |            |                     |            |         |        |  |  |  |  |
| THM-001  |            |                     |            |         |        |  |  |  |  |
| PLD-001  |            |                     |            |         |        |  |  |  |  |
| HRN-001  |            |                     |            |         |        |  |  |  |  |
| Budget   | BOM Total  | Parametric Estimate | Allocation | Margin  | Margin |  |  |  |  |
| Dry mass | _____ g    | _____ g             | _____ g    | _____ g | _____  |  |  |  |  |
| Peak     | _____ W    | _____ W             | _____ W    | _____ W | _____  |  |  |  |  |
| Total    | _____ kEUR | _____ kEUR          | _____      | _____   | _____  |  |  |  |  |
| Volume   | _____ % of |                     | 100%       | _____ % | _____  |  |  |  |  |

\_\_\_\_U

Does everything fit within allocation? Y / N

If not, what would you change? \_\_\_\_\_

## Part D: Export Control Assessment

List any components that may require export licences:

| Component | ECCN / USML Category | Controlled? | Licence Required? | Action Needed |
|-----------|----------------------|-------------|-------------------|---------------|
|           |                      | Y / N       | Y / N             |               |
|           |                      | Y / N       | Y / N             |               |
|           |                      | Y / N       | Y / N             |               |
|           |                      | Y / N       | Y / N             |               |

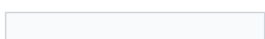
Are there any ITAR-controlled items? Y / N

If yes, what TAA (Technical Assistance Agreement) or DSP-5 licence is needed?

## Part E: Interface Compatibility Check

For each selected component, verify interface compatibility:

| Component | Bus Voltage OK? | Data Protocol OK? | RF Band Match? | Connector OK? | Notes |
|-----------|-----------------|-------------------|----------------|---------------|-------|
|           | Y / N           | Y / N             | Y / N / NA     | Y / N         |       |
|           | Y / N           | Y / N             | Y / N / NA     | Y / N         |       |
|           | Y / N           | Y / N             | Y / N / NA     | Y / N         |       |
|           |                 |                   |                |               |       |



Y / N

Y / N

Y / N / NA

Y / N



|  |       |       |            |       |  |
|--|-------|-------|------------|-------|--|
|  | Y / N | Y / N | Y / N / NA | Y / N |  |
|  | Y / N | Y / N | Y / N / NA | Y / N |  |

## Part F: Interface Incompatibility Resolution

Describe one interface incompatibility found during selection and how you resolved it:

Issue: \_\_\_\_\_

Components involved: \_\_\_\_\_

Root cause: \_\_\_\_\_

Resolution chosen: \_\_\_\_\_

Impact on budget:

## Notes & Reflections

What was the most difficult equipment selection decision? Why?

## Worksheet 4.2: Verification & Validation Planning

Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

### Part A: IADT Method Assignment

For 10 key requirements from your SpaceCDF V&V Matrix, assign the verification method and justify:

| #  | Req ID | Requirement Text (abbreviated) | Method (I/A/D/T) | Phase (B/C/D) | Level (Unit/Sub/Sys) | Justification |
|----|--------|--------------------------------|------------------|---------------|----------------------|---------------|
| 1  |        |                                |                  |               |                      |               |
| 2  |        |                                |                  |               |                      |               |
| 3  |        |                                |                  |               |                      |               |
| 4  |        |                                |                  |               |                      |               |
| 5  |        |                                |                  |               |                      |               |
| 6  |        |                                |                  |               |                      |               |
| 7  |        |                                |                  |               |                      |               |
| 8  |        |                                |                  |               |                      |               |
| 9  |        |                                |                  |               |                      |               |
| 10 |        |                                |                  |               |                      |               |

Summary count: Analysis: \_\_\_\_\_

### Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis (early confidence) AND test (final proof):

### Part C: Environmental Test Sequence

For your selected launch vehicle, specify the complete proto-flight test sequence:

Launch vehicle: \_\_\_\_\_

| Requirement ID | Requirement Text | Analysis (Phase B) | Test (Phase C/D) | Why Both Needed? |
|----------------|------------------|--------------------|------------------|------------------|
|----------------|------------------|--------------------|------------------|------------------|

| Step | Test                      | Specification                    | Duration           | Pass Criteria              |
|------|---------------------------|----------------------------------|--------------------|----------------------------|
| 1    | Initial functional test   | Full performance baseline        | _____ hr           | All parameters nominal     |
| 2    | Sine vibration            | _____ g, _____ - _____ Hz        | _____ min/axis x 3 | No resonance shift > 5%    |
| 3    | Random vibration          | _____ gRMS overall               | _____ min/axis x 3 | No structural damage       |
| 4    | Post-vibe functional test | Same as step 1                   | _____ hr           | Compare to baseline        |
| 5    | Thermal vacuum -- hot     | + _____ C (predicted max + 10 C) | _____ cycles       | Functional at hot extreme  |
| 6    | Thermal vacuum -- cold    | - _____ C (predicted min - 10 C) | _____ cycles       | Functional at cold extreme |
| 7    | Post-TVAC functional test | Same as step 1                   | _____ hr           | Compare to baseline        |
| 8    | EMC test                  | Required? Y / N                  | _____ day          | No interference            |
| 9    | Mass measurement          | Calibrated scale (+/- 0.01 kg)   | 30 min             | M <= _____ kg              |
| 10   | Deployer fit check        | _____ deployer                   | 2 hr               | Fits; slides freely        |

Total test campaign duration estimate: \_\_\_\_\_

## Part D: Vibration Test Profile

Record the random vibration PSD profile from your launch vehicle PUG:

| Frequency (Hz) | PSD Level (g <sup>2</sup> /Hz) | Slope          |
|----------------|--------------------------------|----------------|
| 20             |                                | Start          |
| 20 - _____     |                                | + _____ dB/oct |
| _____ - _____  |                                | Flat           |
| _____ - 2000   |                                | - _____ dB/oct |
| 2000           |                                | End            |

Overall gRMS (calculated): \_\_\_\_\_

Show your calculation:

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## Part E: TVAC Cycle Profile

Sketch or describe one TVAC cycle:

Hot extreme: + \_\_\_\_\_

Ramp rate: \_\_\_\_\_

Functional test at each extreme? Y / N    Duration of functional test: \_\_\_\_\_

Calculate time for one cycle:

t\_cycle = \_\_\_\_\_

= \_\_\_\_\_

Total TVAC campaign: \_\_\_\_\_

## Part F: Waiver Identification

Are there any requirements that cannot be verified by the standard method? Document potential waivers:

\_\_\_\_\_

## Notes & Reflections

Which environmental test do you consider most critical for your mission? Why?

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Worksheet 4.3: Risk, Interfaces & FMECA

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Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

\_\_\_\_\_

## Part A: Risk Register

Identify at least 5 technical risks for your mission design. Score each on the 5x5 matrix.

Scoring reminder: L x C = Score. Low (1-4), Medium (5-9), High (10-15), Critical (16-25).

| Req ID | Standard Method  | Proposed Alternative | Justification | Risk  |                    |                     |                  |                       |
|--------|------------------|----------------------|---------------|-------|--------------------|---------------------|------------------|-----------------------|
| #      | Risk Description | L (1-5)              | C (1-5)       | Score | Category (L/M/H/C) | Mitigation Strategy | Owner (Position) | Post-Mitigation Score |
| 1      |                  |                      |               |       |                    |                     |                  |                       |
| 2      |                  |                      |               |       |                    |                     |                  |                       |
| 3      |                  |                      |               |       |                    |                     |                  |                       |
| 4      |                  |                      |               |       |                    |                     |                  |                       |
| 5      |                  |                      |               |       |                    |                     |                  |                       |
| 6      |                  |                      |               |       |                    |                     |                  |                       |
| 7      |                  |                      |               |       |                    |                     |                  |                       |

How many risks are High or Critical (before mitigation)? \_\_\_\_\_

How many remain High or Critical after mitigation? \_\_\_\_\_

## Part B: 5x5 Risk Matrix Plot

Plot your risks on the matrix below. Write the risk number in the appropriate cell.

|   |   | Consequence -> |   |   |   |   |
|---|---|----------------|---|---|---|---|
|   |   | 1              | 2 | 3 | 4 | 5 |
| L | 5 |                |   |   |   |   |
| i | 4 |                |   |   |   |   |
| k | 3 |                |   |   |   |   |
| e | 2 |                |   |   |   |   |
| l | 1 |                |   |   |   |   |

## Part C: N-Squared (N^2) Interface Matrix

Fill in the interface matrix for your mission's subsystems. Use abbreviations:

- P = Power (electrical)
- D = Data (I2C, SPI, UART, CAN, RS-422)
- R = RF (coaxial)
- M = Mechanical (mounting, alignment)
- T = Thermal (conductive/radiative)

|           | EPS | OBC | AOCS | TTC | Payload | Structure |
|-----------|-----|-----|------|-----|---------|-----------|
| EPS       | --- |     |      |     |         |           |
| OBC       |     | --- |      |     |         |           |
| AOCS      |     |     | ---  |     |         |           |
| TTC       |     |     |      | --- |         |           |
| Payload   |     |     |      |     | ---     |           |
| Structure |     |     |      |     |         | ---       |

Total number of interfaces identified: \_\_\_\_\_

Interface conflicts detected: (describe at least 2)

1. Conflict: \_\_\_\_\_

Resolution: \_\_\_\_\_

2. Conflict: \_\_\_\_\_

Resolution: \_\_\_\_\_

## Part D: FMECA Table

Complete for 4 critical components:

## Part E: Single-Point Failure Analysis

List all single-point failures in your design:

| Component | Function  | Failure Mode | Cause             | Local Effect | System Effect | Severity (1-5)                 | Detection Method | Compensating Provision |
|-----------|-----------|--------------|-------------------|--------------|---------------|--------------------------------|------------------|------------------------|
| #         | Component | Failure Mode | Effect on Mission | SPF?         | Acceptable?   | Mitigation (if not acceptable) |                  |                        |
| 1         |           |              |                   | Y/N          | Y/N           |                                |                  |                        |
| 2         |           |              |                   | Y/N          | Y/N           |                                |                  |                        |
| 3         |           |              |                   | Y/N          | Y/N           |                                |                  |                        |
| 4         |           |              |                   | Y/N          | Y/N           |                                |                  |                        |
| 5         |           |              |                   | Y/N          | Y/N           |                                |                  |                        |

Total SPFs identified: \_\_\_\_\_

## Part F: Reliability Calculation

Calculate the series reliability for your mission's critical chain (components that must all work for mission success):

**R<sub>series</sub>** =  $R_1 \times R_2 \times \dots \times R_n$  = \_\_\_\_\_

**R<sub>series</sub>** = \_\_\_\_\_

If you added redundancy to one component (e.g., dual deployment mechanism), recalculate:

**R<sub>redundant</sub>** =  $1 - (1 - R)^2 = 1 - (1 -$  \_\_\_\_\_

**New R<sub>series</sub>** (with redundancy) = \_\_\_\_\_

**Improvement:** from \_\_\_\_\_

## Part G: Risk Discussion

What is the highest-risk item in your design? What would it take (cost, mass, schedule) to bring the risk to an acceptable level?

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## Notes & Reflections

Which single-point failure concerns you the most? Would you accept it, or is mitigation essential?

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## Worksheet 4.4: Cost Estimation & Design Review

Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

### Part A: Mission Cost Estimate (WBS)

Complete using both parametric estimates (from SpaceCDF Cost tab) and bottom-up estimates (from BOM vendor quotes):

| Compon | Individual Reliability (R <sub>i</sub> ) | Mission Duration Used |                  |            |       |
|--------|------------------------------------------|-----------------------|------------------|------------|-------|
| WBS #  | Element                                  | Parametric (kEUR)     | Bottom-Up (kEUR) | Difference | Notes |
| 1.0    | Programme Management                     |                       |                  |            |       |
| 2.0    | Systems Engineering                      |                       |                  |            |       |
| 3.0    | Mission Assurance                        |                       |                  |            |       |
| 4.0    | Payload                                  |                       |                  |            |       |
| 5.0    | Spacecraft Bus Hardware                  |                       |                  |            |       |
|        | 5.1 Structure & mechanisms               |                       |                  |            |       |
|        | 5.2 EPS                                  |                       |                  |            |       |
|        | 5.3 AOCS                                 |                       |                  |            |       |
|        | 5.4 TTC                                  |                       |                  |            |       |
|        | 5.5 OBC & data handling                  |                       |                  |            |       |
|        | 5.6 Thermal                              |                       |                  |            |       |
|        | 5.7 Propulsion                           |                       |                  |            |       |
|        | 5.8 Harness                              |                       |                  |            |       |
| 6.0    | Integration & Test                       |                       |                  |            |       |
| 7.0    | Software (FSW + GSW)                     |                       |                  |            |       |
| 8.0    | Launch Services                          |                       |                  |            |       |
| 9.0    | Ground Segment                           |                       |                  |            |       |
| 10.0   | Operations (____ years)                  |                       |                  |            |       |
|        | TOTAL                                    | _____                 | _____            |            |       |

### Part B: Confidence Levels

P50 estimate (your best estimate): \_\_\_\_\_

P70 estimate (P50 x 1.2): \_\_\_\_\_

P80 estimate (P50 x 1.3): \_\_\_\_\_

Show your reasoning for the multiplier (is 1.3 appropriate, or should it be higher for your mission?):

\_\_\_\_\_

## Part C: Constellation Cost (if applicable)

Number of satellites: \_\_\_\_\_

**\*\*b** =  $\ln(\text{learning\_rate}) / \ln(2) = \ln($  \_\_\_\_\_

| Unit #        | Unit Cost (kEUR) = $C_1 \times N^b$ | Cumulative (kEUR) |
|---------------|-------------------------------------|-------------------|
| 1             |                                     |                   |
| 2             |                                     |                   |
| 4             |                                     |                   |
| 8             |                                     |                   |
| _____ (total) |                                     |                   |

Show your calculation for the total constellation cost:

Hardware: \_\_\_\_\_

Spares ( \_\_\_\_\_

Launch: \_\_\_\_\_

Ground segment: \_\_\_\_\_

Operations ( \_\_\_\_\_

PM/SE/MA: \_\_\_\_\_

**\*\*Total constellation cost:** \_\_\_\_\_

**\*\*P80 constellation cost (x 1.3):** \_\_\_\_\_

## Part D: Cost Drivers

Top 3 cost drivers in your mission (from SpaceCDF Cost Breakdown):

| Rank | Cost Element | Cost (kEUR) | % of Total | Why is it Expensive? |
|------|--------------|-------------|------------|----------------------|
| 1    |              |             |            |                      |
| 2    |              |             |            |                      |
| 3    |              |             |            |                      |

What would you change to reduce total cost by 20%?

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## Part E: Design Review Readiness

From the SpaceCDF Gate Review tab, assess your readiness for SRR/PDR:

| #  | Gate Criterion                         | Status (Pass/Fail/Manual) | Evidence Location | If Fail: Action Needed |
|----|----------------------------------------|---------------------------|-------------------|------------------------|
| 1  | Mission need justified                 |                           |                   |                        |
| 2  | Requirements complete & traceable      |                           |                   |                        |
| 3  | All budgets close with margin          |                           |                   |                        |
| 4  | Equipment selected (BOM complete)      |                           |                   |                        |
| 5  | Interfaces defined, conflicts resolved |                           |                   |                        |
| 6  | V&V approach defined                   |                           |                   |                        |
| 7  | Risks identified with mitigations      |                           |                   |                        |
| 8  | Cost estimate with WBS                 |                           |                   |                        |
| 9  | ConOps defined                         |                           |                   |                        |
| 10 | Sustainability compliance              |                           |                   |                        |

Must-pass criteria met: \_\_\_\_\_

Overall readiness: READY / NOT READY

Action items to achieve readiness:

1. \_\_\_\_\_
2. \_\_\_\_\_
3. \_\_\_\_\_

## Part F: Presentation Preparation

Outline your 12-minute design review presentation:

| Section                 | Duration | Key Message | Evidence/Data to Show |
|-------------------------|----------|-------------|-----------------------|
| 1. Mission Need         | 2 min    |             |                       |
| 2. Why Space?           | 2 min    |             |                       |
| 3. System Design        | 3 min    |             |                       |
| 4. Equipment & V&V      | 2 min    |             |                       |
| 5. Risk & Cost          | 2 min    |             |                       |
| 6. Operations & Lessons | 1 min    |             |                       |

## Notes & Reflections

What is the most uncertain element in your cost estimate? How would you reduce the uncertainty?

## Worksheet 5.1: Ground Segment & Operations Architecture

Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

### Part A: Ground Station Contact Analysis

Calculate contact time and data volume per pass for your mission.

#### Input Parameters

| Parameter                         | Value                       | Source                               |
|-----------------------------------|-----------------------------|--------------------------------------|
| Orbital altitude (h)              | _____ km                    | SpaceCDF orbit selection             |
| Minimum elevation angle (epsilon) | _____ deg                   | Ground station spec (typically 5-10) |
| Downlink data rate (R)            | _____ Mbps                  | SpaceCDF link budget                 |
| Protocol efficiency (eta)         | _____ (typically 0.85-0.95) | CCSDS overhead estimate              |
| Ground station latitude           | _____ deg                   | Selected station                     |
| Earth radius (R_E)                | 6371 km                     | Constant                             |

#### Calculations

Maximum slant range:

$$\rho = R_E \times (\sqrt{(h/R_E + 1)^2 - \cos^2(\epsilon)} - \sin(\epsilon))$$

$$\rho = 6371 \times (\sqrt{((\text{_____})^2 - \cos^2(\epsilon))} - \sin(\epsilon))$$

$$\rho = \text{_____}$$

Average pass duration (approximate):

$$T_{\text{avg}} \sim \text{_____}$$

Data volume per pass:

$$V_{\text{data}} = R \times T \times \eta = \text{_____}$$

## Part B: Data Budget Closure

| Parameter                            | Value | Unit   |
|--------------------------------------|-------|--------|
| Daily data generation (payload)      |       | MB/day |
| Daily data generation (housekeeping) |       | MB/day |
| Total daily generation               |       | MB/day |
| Passes per day (GS 1: _____)         |       | passes |
| Data per pass (GS 1)                 |       | MB     |
| Subtotal downlink (GS 1)             |       | MB/day |
| Passes per day (GS 2: _____)         |       | passes |
| Data per pass (GS 2)                 |       | MB     |
| Subtotal downlink (GS 2)             |       | MB/day |
| Total daily downlink capacity        |       | MB/day |
| Data budget margin                   |       | MB/day |
| Margin percentage                    |       | %      |

Does the data budget close? Y / N

If not, what changes would close it? (check all that apply)

- ☐ Add another ground station at: \_\_\_\_\_
- ☐ Increase data rate to: \_\_\_\_\_ Mbps (requires: \_\_\_\_\_)
- ☐ Reduce payload data generation to: \_\_\_\_\_ MB/day (impact: \_\_\_\_\_)
- ☐ Add onboard compression (ratio: \_\_\_\_\_:1)
- ☐ Use SatNOGS network for additional passes
- ☐ Other: \_\_\_\_\_

## Part C: Ground Station Network Design

| Station # | Location | Latitude | Antenna | Band | Data Rate | Passes/Day | Role           |
|-----------|----------|----------|---------|------|-----------|------------|----------------|
| 1         |          |          |         |      |           |            | Primary TTC    |
| 2         |          |          |         |      |           |            | Payload DL     |
| 3         |          |          |         |      |           |            | Backup/SatNOGS |

Total ground segment cost estimate: \_\_\_\_\_

| Cost Element                 | Annual Cost (kEUR) | Notes |
|------------------------------|--------------------|-------|
| Antenna rental / ownership   |                    |       |
| MCS software licence         |                    |       |
| Network connectivity         |                    |       |
| Operations staff (_____ FTE) |                    |       |
| Total annual                 |                    |       |
| Mission lifetime (_____ yr)  |                    |       |

## Part D: LEOP Timeline

Construct the LEOP timeline for the first 72 hours after separation:

| Time (UTC) | Sim Time      | Activity                     | Success Criterion             | Status |
|------------|---------------|------------------------------|-------------------------------|--------|
| T+0        | Separation    | Deployment switches release  |                               | []     |
| T+30 min   | Timer expires | Antenna deployment commanded |                               | []     |
| T+_____    |               | Beacon acquisition           | Carrier lock                  | []     |
| T+_____    |               | First HK telemetry           | Data decoded                  | []     |
| T+_____    |               | First uplink command         | ACK received                  | []     |
| T+_____    |               | SA deployment (if separate)  | Power generation confirmed    | []     |
| T+_____    |               | ADCS initialisation          | Attitude determination active | []     |
| T+_____    |               | ADCS calibration             | Pointing < _____ deg          | []     |
| T+_____    |               | Full duplex validation       | Uplink + downlink at ops rate | []     |
| T+_____    |               | Health assessment complete   | All subsystems nominal        | []     |

## Part E: Mission Operations Timeline (Gantt Chart)

Sketch the operations timeline from launch to end of life:

|           |       |   |   |            |   |   |           |   |   |                    |    |     |     |     |     |     |     |
|-----------|-------|---|---|------------|---|---|-----------|---|---|--------------------|----|-----|-----|-----|-----|-----|-----|
| Phase:    | LEOP  |   |   | Commission |   |   | Early Ops |   |   | Nominal Operations |    |     | EOL |     |     |     |     |
| Week:     | 1     | 2 | 3 | 4          | 5 | 6 | 7         | 8 | 9 | ...                | 26 | ... | 52  | ... | 104 | ... | 156 |
|           |       |   |   |            |   |   |           |   |   |                    |    |     |     |     |     |     |     |
| Staffing: | _____ |   |   |            |   |   |           |   |   |                    |    |     |     |     |     |     |     |
|           | 24/7  |   |   | 16/7       |   |   | 8/5       |   |   | 8/5 or automated   |    |     | 8/5 |     |     |     |     |

Key milestones:

- L+\_\_: First light
- L+\_\_: Commissioning complete
- L+\_\_: Orbit maintenance manoeuvre (if applicable)
- L+\_\_: End of nominal mission
- L+\_\_: Extended mission (if applicable)
- L+\_\_: Passivation and deorbit

## Notes & Reflections

What is the most challenging aspect of your ground segment design? What would you do differently with more budget?

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## Worksheet 5.2: Mission Operations Concepts

Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

### Part A: Operational Modes

Define all operational modes for your spacecraft. Use the SpaceCDF ConOps Editor to enter these.

Verify: Does the power budget close in every mode?

| Mode                   | Description           | Power (W)           | Data Rate | Entry Condition | Exit Condition |
|------------------------|-----------------------|---------------------|-----------|-----------------|----------------|
| Safe                   |                       |                     |           |                 |                |
| Detumble               |                       |                     |           |                 |                |
| Standby                |                       |                     |           |                 |                |
| Science                |                       |                     |           |                 |                |
| Downlink               |                       |                     |           |                 |                |
| Eclipse                |                       |                     |           |                 |                |
| Manoeuvre              |                       |                     |           |                 |                |
| Mode                   | Power Consumption (W) | Power Available (W) | Margin    | Closes?         |                |
| Safe                   |                       |                     |           | Y / N           |                |
| Detumble               |                       |                     |           | Y / N           |                |
| Standby                |                       |                     |           | Y / N           |                |
| Science                |                       |                     |           | Y / N           |                |
| Downlink               |                       |                     |           | Y / N           |                |
| Eclipse (battery only) |                       |                     |           | Y / N           |                |

### Part B: FDIR Rules

Define at least 5 FDIR rules for your spacecraft:

| # | Fault Description | Detection Method | Thresho | FDIR Level (0-4) | Autonomous | Recovery |
|---|-------------------|------------------|---------|------------------|------------|----------|
| 1 |                   |                  |         |                  |            |          |
| 2 |                   |                  |         |                  |            |          |
| 3 |                   |                  |         |                  |            |          |
| 4 |                   |                  |         |                  |            |          |
| 5 |                   |                  |         |                  |            |          |
| 6 |                   |                  |         |                  |            |          |

Can Safe Mode sustain the spacecraft indefinitely (power + thermal)? Y / N

If not, what is the maximum Safe Mode duration? \_\_\_\_\_

Can every autonomous FDIR action be overridden from ground? Y / N

If not, which cannot? \_\_\_\_\_

## Part C: Nominal Procedure

Write one nominal procedure for your mission (science observation or data downlink):

Procedure ID: \_\_\_\_\_

Preconditions:

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| Step | Action | Expected Result | If Not Achieved |
|------|--------|-----------------|-----------------|
| 1    |        |                 |                 |
| 2    |        |                 |                 |
| 3    |        |                 |                 |
| 4    |        |                 |                 |
| 5    |        |                 |                 |
| 6    |        |                 |                 |
| 7    |        |                 |                 |
| 8    |        |                 |                 |

Post-conditions:

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## Part D: Contingency Procedure

Write one contingency procedure (e.g., safe mode recovery, communication loss recovery):

Procedure ID: \_\_\_\_\_

Trigger condition:

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Severity level: Green / Yellow / Orange / Red

| Step | Action | Expected Result | If Not Achieved | Timeout |
|------|--------|-----------------|-----------------|---------|
| 1    |        |                 |                 |         |
| 2    |        |                 |                 |         |
| 3    |        |                 |                 |         |
| 4    |        |                 |                 |         |
| 5    |        |                 |                 |         |
| 6    |        |                 |                 |         |

Escalation: If this procedure fails, escalate to: \_\_\_\_\_

Post-conditions:

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## Part E: Operations Staffing

Estimate the staffing requirements for each mission phase:

| Phase              | Duration     | Shifts/Day      | Staff/Shift | Total FTE | Annual Cost (kEUR) |
|--------------------|--------------|-----------------|-------------|-----------|--------------------|
| LEOP               | _____ days   | 3 (24/7)        |             |           |                    |
| Commissioning      | _____ weeks  | 2 (16/7)        |             |           |                    |
| Early Operations   | _____ months | 1 (8/5)         |             |           |                    |
| Nominal Operations | _____ years  | 1 (8/5) or auto |             |           |                    |
| End of Life        | _____ weeks  | 1 (8/5)         |             |           |                    |
| Total              |              |                 |             |           |                    |

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## Part F: Anomaly Scenario

Hypothetical scenario: Your satellite enters Safe Mode during a weekend. The next ground contact is Monday morning (40 hours away). Walk through your response:

1. What is the spacecraft doing in Safe Mode?

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2. Will it survive 40 hours? (Check power and thermal)

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3. What telemetry will you review first on Monday?

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4. What is your recovery plan?

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## Notes & Reflections

How much autonomy should your spacecraft have? Where is the line between onboard decision-making and ground control?

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## Worksheet 5.3: Mission Simulation Day

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Name: 

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Mission Name: 

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Your Role: 

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## Simulation Event Log

Record ALL events, decisions, and actions in real-time during the simulation:

| #  | Sim Time | Real Time | Event / Observation | Action Taken | Result | Logged By |
|----|----------|-----------|---------------------|--------------|--------|-----------|
| 1  | T+0      |           | Separation          |              |        |           |
| 2  |          |           |                     |              |        |           |
| 3  |          |           |                     |              |        |           |
| 4  |          |           |                     |              |        |           |
| 5  |          |           |                     |              |        |           |
| 6  |          |           |                     |              |        |           |
| 7  |          |           |                     |              |        |           |
| 8  |          |           |                     |              |        |           |
| 9  |          |           |                     |              |        |           |
| 10 |          |           |                     |              |        |           |
| 11 |          |           |                     |              |        |           |
| 12 |          |           |                     |              |        |           |
| 13 |          |           |                     |              |        |           |
| 14 |          |           |                     |              |        |           |
| 15 |          |           |                     |              |        |           |
| 16 |          |           |                     |              |        |           |
| 17 |          |           |                     |              |        |           |
| 18 |          |           |                     |              |        |           |
| 19 |          |           |                     |              |        |           |
| 20 |          |           |                     |              |        |           |

## LEOP Checklist

Check off each LEOP milestone as it is achieved during the simulation:

| #  | Milestone                      | Time Achieved | Status                   | Notes |
|----|--------------------------------|---------------|--------------------------|-------|
| 1  | Separation confirmed           |               | <input type="checkbox"/> |       |
| 2  | 30-min timer expired           |               | <input type="checkbox"/> |       |
| 3  | Antenna deployed               |               | <input type="checkbox"/> |       |
| 4  | Beacon acquired                |               | <input type="checkbox"/> |       |
| 5  | First HK telemetry decoded     |               | <input type="checkbox"/> |       |
| 6  | First uplink command ACK'd     |               | <input type="checkbox"/> |       |
| 7  | Solar array deployed (if sep.) |               | <input type="checkbox"/> |       |
| 8  | ADCS detumble complete         |               | <input type="checkbox"/> |       |
| 9  | ADCS calibration complete      |               | <input type="checkbox"/> |       |
| 10 | Full duplex validated          |               | <input type="checkbox"/> |       |
| 11 | All subsystems nominal         |               | <input type="checkbox"/> |       |

## Anomaly Response Form 1

Complete this form for each injected anomaly during the simulation.

| Field      | Entry       |
|------------|-------------|
| Anomaly ID | ANO-SIM-001 |
| Sim Time   |             |
| Real Time  |             |

|                    |                               |
|--------------------|-------------------------------|
| Severity           | Green / Yellow / Orange / Red |
| Affected Subsystem |                               |

Observation (exact telemetry values and symptoms):

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Initial Assessment (what do you think happened?):

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Diagnosis (root cause after investigation):

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Action Taken (commands sent, procedures executed):

| Step | Time | Action | Expected Result | Actual Result |
|------|------|--------|-----------------|---------------|
| 1    |      |        |                 |               |
| 2    |      |        |                 |               |
| 3    |      |        |                 |               |
| 4    |      |        |                 |               |
| 5    |      |        |                 |               |

Final Result (system state after response):

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Follow-up Required:

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Procedure used: \_\_\_\_\_

If not, what changes are needed? \_\_\_\_\_

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## Anomaly Response Form 2

| Field              | Entry                         |
|--------------------|-------------------------------|
| Anomaly ID         | ANO-SIM-002                   |
| Sim Time           |                               |
| Real Time          |                               |
| Severity           | Green / Yellow / Orange / Red |
| Affected Subsystem |                               |

Observation (exact telemetry values and symptoms):

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Initial Assessment (what do you think happened?):

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Diagnosis (root cause after investigation):

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Action Taken (commands sent, procedures executed):

| Step | Time | Action | Expected Result | Actual Result |
|------|------|--------|-----------------|---------------|
| 1    |      |        |                 |               |
| 2    |      |        |                 |               |
| 3    |      |        |                 |               |
| 4    |      |        |                 |               |
| 5    |      |        |                 |               |

Final Result (system state after response):

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Follow-up Required:

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Procedure used:

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If not, what changes are needed? \_\_\_\_\_

\_\_\_\_\_

## Anomaly Response Form 3

| Field              | Entry                         |
|--------------------|-------------------------------|
| Anomaly ID         | ANO-SIM-003                   |
| Sim Time           |                               |
| Real Time          |                               |
| Severity           | Green / Yellow / Orange / Red |
| Affected Subsystem |                               |

Observation (exact telemetry values and symptoms):

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Initial Assessment:

\_\_\_\_\_

\_\_\_\_\_

Diagnosis:

\_\_\_\_\_

\_\_\_\_\_

Action Taken:

| Step | Time | Action | Expected Result | Actual Result |
|------|------|--------|-----------------|---------------|
| 1    |      |        |                 |               |
| 2    |      |        |                 |               |
| 3    |      |        |                 |               |
| 4    |      |        |                 |               |

Final Result:

\_\_\_\_\_

\_\_\_\_\_

Procedure used: \_\_\_\_\_

\_\_\_\_\_

## Commissioning Status

After the commissioning phase of the simulation, record subsystem checkout status:

| Subsystem  | Checkout Complete? | Performance vs Spec              | Issues Found |
|------------|--------------------|----------------------------------|--------------|
| EPS        | [ ]                | Nominal / Degraded / Failed      |              |
| OBC        | [ ]                | Nominal / Degraded / Failed      |              |
| AOCS       | [ ]                | Nominal / Degraded / Failed      |              |
| TTC        | [ ]                | Nominal / Degraded / Failed      |              |
| Thermal    | [ ]                | Nominal / Degraded / Failed      |              |
| Payload    | [ ]                | Nominal / Degraded / Failed      |              |
| Propulsion | [ ]                | Nominal / Degraded / Failed / NA |              |

## Nominal Operations Summary

After the nominal operations phase, record:

| Metric                         | Planned | Actual (Simulated) | Variance |
|--------------------------------|---------|--------------------|----------|
| Science observations completed |         |                    |          |
| Data downlinked (MB)           |         |                    |          |
| Anomalies encountered          |         |                    |          |
| Ground contacts made           |         |                    |          |
| Safe mode entries              |         |                    |          |

## Simulation Debrief

**What went well?**

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**What could be improved?**

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**Design weaknesses revealed by simulation:**

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**Procedure changes needed:**

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**FDIR gaps identified:**

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**Notes & Reflections**

What was the most stressful moment during the simulation? What did it teach you about operations?

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## Worksheet 5.4: Final Review & Presentations

Name: \_\_\_\_\_

Mission Name: \_\_\_\_\_

### Part A: Final Design Summary

Record the final state of your design from the SpaceCDF Dashboard:

| Parameter                | Value | Unit   | Margin | Margin % | Status (G/A/R) |
|--------------------------|-------|--------|--------|----------|----------------|
| Dry mass                 |       | kg     |        |          |                |
| Wet mass (if propulsion) |       | kg     |        |          |                |
| SA power (EOL)           |       | W      |        |          |                |
| Battery capacity         |       | Wh     |        |          |                |
| Link margin (worst case) |       | dB     |        |          |                |
| Pointing accuracy        |       | deg    |        |          |                |
| Data balance             |       | MB/day |        |          |                |
| Total cost (P50)         |       | kEUR   |        |          |                |
| Total cost (P80)         |       | kEUR   |        |          |                |
| Debris compliance score  |       | /100   |        |          |                |
| Mission lifetime         |       | years  |        |          |                |

Do all budgets close? Y / N    If not, which fail? \_\_\_\_\_

Number of unresolved conflicts: \_\_\_\_\_

### Part B: Documentation Checklist

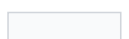
Tick each document generated from SpaceCDF Exports tab:

| Document                              | Generated?               | Complete?                | Notes |
|---------------------------------------|--------------------------|--------------------------|-------|
| MRD (Mission Requirements Document)   | <input type="checkbox"/> | <input type="checkbox"/> |       |
| Technical Specification               | <input type="checkbox"/> | <input type="checkbox"/> |       |
| Verification Plan                     | <input type="checkbox"/> | <input type="checkbox"/> |       |
| ConOps Document                       | <input type="checkbox"/> | <input type="checkbox"/> |       |
| SEMP (Systems Engineering Mgmt Plan)  | <input type="checkbox"/> | <input type="checkbox"/> |       |
| IRD (Interface Requirements Document) | <input type="checkbox"/> | <input type="checkbox"/> |       |
| Risk Management Plan                  | <input type="checkbox"/> | <input type="checkbox"/> |       |
| Test Plan                             | <input type="checkbox"/> | <input type="checkbox"/> |       |
| Bill of Materials                     | <input type="checkbox"/> | <input type="checkbox"/> |       |
| ITU Filing Template                   | <input type="checkbox"/> | <input type="checkbox"/> |       |
| RSSSA Filing                          | <input type="checkbox"/> | <input type="checkbox"/> |       |

Export Control Assessment

[]

[]





|                     |                          |                          |  |
|---------------------|--------------------------|--------------------------|--|
| End-of-Life Report  | <input type="checkbox"/> | <input type="checkbox"/> |  |
| COPUOS Registration | <input type="checkbox"/> | <input type="checkbox"/> |  |

Total documents generated: \_\_\_\_\_

## Part C: Presentation Outline

Prepare your 12-minute final design review presentation:

| Section                 | Time  | Key Points to Cover | Evidence / Data to Show |
|-------------------------|-------|---------------------|-------------------------|
| 1. Mission Need         | 2 min |                     |                         |
| 2. Why Space?           | 2 min |                     |                         |
| 3. System Design        | 3 min |                     |                         |
| 4. Equipment & V&V      | 2 min |                     |                         |
| 5. Risk & Cost          | 2 min |                     |                         |
| 6. Operations & Lessons | 1 min |                     |                         |

Presenter(s): \_\_\_\_\_

Backup slides prepared on: \_\_\_\_\_

## Part D: Peer Review Evaluation

Complete this section while reviewing another team's presentation.

Team being reviewed: \_\_\_\_\_

Reviewers: \_\_\_\_\_

| #  | Criterion                                | Score (0-2) | Comments / Questions |
|----|------------------------------------------|-------------|----------------------|
| 1  | Mission need clearly justified           |             |                      |
| 2  | Requirements complete and traceable      |             |                      |
| 3  | All budgets close with margin            |             |                      |
| 4  | Equipment selected with justified trades |             |                      |
| 5  | Interfaces defined, conflicts resolved   |             |                      |
| 6  | V&V approach defined                     |             |                      |
| 7  | Risks identified with mitigations        |             |                      |
| 8  | Cost estimate with WBS and P80           |             |                      |
| 9  | ConOps and ground segment designed       |             |                      |
| 10 | Sustainability compliance                |             |                      |
|    | TOTAL                                    | / 20        |                      |

Presentation quality:

| Criterion                              | Score (1-5) | Comments |
|----------------------------------------|-------------|----------|
| Clarity of communication               |             |          |
| Use of evidence (data, not assertions) |             |          |
| Ability to answer questions            |             |          |
| Time management                        |             |          |

Board recommendation: GO / NO GO / GO WITH ACTIONS

Action items for the presenting team:

| # | Action | Responsible | Deadline |
|---|--------|-------------|----------|
| 1 |        |             |          |
| 2 |        |             |          |
| 3 |        |             |          |

Strongest aspect of this team's design:

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Biggest area for improvement:

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## Part E: Lessons Learned

1. What was the most challenging design decision your team faced?

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2. What would you do differently if starting the course over?

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3. What does the design need next (Phase B activities)?

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4. What was the most valuable SpaceCDF feature?

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5. What feature was missing that you needed?

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6. What concept from this course will you apply in your future work?

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## Part F: Self-Assessment Rubric

Rate your own learning across the three weeks of the course. Be honest -- this is for your own reflection, not grading.

| Competency                                                         | Week 1 Start (1-5) | Now (1-5) | Evidence of Growth |
|--------------------------------------------------------------------|--------------------|-----------|--------------------|
| Define a space mission starting from the problem, not the solution |                    |           |                    |
| Write verifiable, traceable requirements (SMART format)            |                    |           |                    |
| Conduct a structured trade study with weighted criteria            |                    |           |                    |
| Size subsystems using parametric models and engineering budgets    |                    |           |                    |
| Select COTS equipment and verify interface compatibility           |                    |           |                    |
| Construct a verification matrix with IADT methods                  |                    |           |                    |
| Assess risk using a 5x5 matrix and define mitigations              |                    |           |                    |
| Estimate mission cost using parametric and bottom-up methods       |                    |           |                    |
| Design a ground segment and calculate data budget                  |                    |           |                    |
| Define operational modes, FDIR rules, and procedures               |                    |           |                    |
| Respond to anomalies using a systematic process                    |                    |           |                    |
| Present a design to a review board with evidence                   |                    |           |                    |

Scale: 1 = No familiarity, 2 = Basic awareness, 3 = Can apply with guidance, 4 = Can apply independently, 5 = Can teach others

Overall self-assessment:

At the start of this course, I would describe my systems engineering ability as:

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Now, I would describe it as:

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One thing I am most proud of from this course:

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## Notes & Reflections

Final thoughts on the course, the design process, and what comes next:

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